

SHIRAKAWA, Japan

WMO: 475970

Lat: 37.1321N

Lon: 140.2150E

Elev: 357

StdP: 97.11

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.8	-4.6	-12.1	1.4	-0.6	-11.0	1.5	-0.5	12.8	1.1	11.7	1.6	2.2	290	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	31.7	24.1	30.3	23.6	28.9	22.9	25.1	29.7	24.4	28.6	23.8	27.4	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.8	19.5	26.9	23.2	18.8	26.5	22.7	18.2	26.0	78.8	29.8	76.2	28.8	73.6	27.5	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.7	8.5	-8.2	33.7	1.1	1.0	-9.0	34.4	-9.6	35.0	-10.2	35.5	-11.0	36.3		
(5)			-8.9	26.0	1.1	0.6	-9.7	26.4	-10.3	26.8	-10.9	27.1	-11.7	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	0.6	1.3	4.6	10.3	15.7	19.4	23.4	24.4	20.3	14.4	8.2	3.2	(6)
(7)		DBStd	8.77	2.32	2.71	3.24	3.64	2.97	2.70	2.95	2.76	3.26	3.03	3.05	2.75	(7)
(8)		HDD10.0	1031	290	244	171	41	2	0	0	0	0	3	71	210	(8)
(9)		HDD18.3	2722	548	477	426	242	89	19	2	1	17	128	304	468	(9)
(10)		CDD10.0	1837	0	0	3	48	178	281	414	447	309	138	17	1	(10)
(11)		CDD18.3	486	0	0	0	0	8	51	158	189	76	4	0	0	(11)
(12)		CDH23.3	3069	0	0	0	8	82	246	1036	1324	361	12	0	0	(12)
(13)		CDH26.7	896	0	0	0	1	11	35	337	443	69	0	0	0	(13)
(14)	Wind	WSAvg	3.5	3.8	4.1	4.4	4.3	3.8	3.0	2.8	2.7	2.9	3.2	3.4	3.6	(14)
(15)	Precipitation	PrecAvg	1437	54	46	75	103	118	143	217	200	200	144	81	57	(15)
(16)		PrecMax	1988	132	112	126	197	240	246	432	715	374	476	204	138	(16)
(17)		PrecMin	1031	19	11	21	14	47	62	63	46	9	19	11	16	(17)
(18)		PrecStd	228	31	23	26	50	48	49	82	129	87	103	45	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	13.7	18.7	24.3	28.3	29.5	33.3	33.6	30.4	25.1	19.2	14.0	(19)
(20)			MCWB	5.9	8.0	10.0	13.8	17.6	20.9	24.1	24.8	23.1	18.9	13.5	9.6	(20)
(21)		2%	DB	8.2	10.4	15.4	21.4	25.6	27.6	31.6	32.0	28.7	22.4	17.0	11.6	(21)
(22)			MCWB	3.9	5.5	8.4	12.5	16.3	20.1	24.0	24.5	22.5	17.4	12.3	7.7	(22)
(23)		5%	DB	6.4	8.3	13.0	19.3	23.7	26.1	30.2	30.7	27.0	20.9	15.4	9.9	(23)
(24)			MCWB	2.5	3.9	7.7	12.0	15.5	19.3	23.6	24.1	21.7	16.5	11.3	6.2	(24)
(25)		10%	DB	5.0	6.4	11.0	17.2	21.9	24.6	28.6	29.3	25.4	19.5	13.9	8.3	(25)
(26)			MCWB	1.6	2.5	6.3	10.8	14.9	18.8	22.9	23.7	21.2	15.5	10.7	5.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	9.2	12.1	17.0	19.6	23.1	25.7	26.1	24.3	20.6	15.1	10.9	(27)
(28)			MCD B	10.4	12.9	15.1	20.0	23.9	26.3	30.5	30.8	28.0	23.1	17.3	12.2	(28)
(29)		2%	WB	4.4	6.1	10.2	14.6	18.2	21.9	24.9	25.3	23.5	18.9	13.6	8.5	(29)
(30)			MCD B	7.4	9.2	14.0	19.2	22.5	25.1	29.6	29.9	26.7	21.1	15.5	10.7	(30)
(31)		5%	WB	3.0	4.3	8.4	13.2	17.1	21.0	24.2	24.8	22.8	17.6	12.4	6.7	(31)
(32)			MCD B	5.9	7.7	12.4	16.7	21.0	24.2	28.4	29.1	25.8	19.8	14.5	9.4	(32)
(33)		10%	WB	1.8	2.8	6.7	11.9	16.1	19.9	23.6	24.2	22.0	16.3	11.0	5.2	(33)
(34)			MCD B	4.6	6.1	10.5	15.6	20.0	23.0	27.3	28.0	24.6	18.8	13.5	8.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.1	9.3	10.3	9.9	8.3	7.5	7.8	7.7	8.4	9.0	8.0	(35)
(36)			MCD BR	10.0	11.6	13.4	13.9	13.4	11.0	10.3	9.9	9.1	9.5	11.2	10.4	(36)
(37)			MCWBR	6.6	7.7	8.3	7.1	6.1	4.9	4.0	3.8	4.1	5.3	7.2	7.1	(37)
(38)		5% WB	MCD BR	9.6	11.1	11.8	10.4	10.2	8.2	9.0	8.9	7.9	7.8	9.3	9.9	(38)
(39)			MCWBR	6.8	7.9	8.4	6.7	5.8	4.2	3.9	3.8	3.9	5.1	6.8	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.353	0.398	0.468	0.519	0.540	0.570	0.573	0.568	0.489	0.436	0.386	0.354	(40)	
(41)		taud	2.217	2.076	1.935	1.872	1.871	1.865	1.920	1.927	2.102	2.170	2.237	2.269	(41)	
(42)		Ebn at Noon	812	813	789	772	763	739	733	725	763	767	767	782	(42)	
(43)		Edh at Noon	106	138	175	197	201	202	190	185	147	125	104	95	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.27	2.97	3.85	4.63	5.14	4.77	4.58	4.58	3.71	2.97	2.29	1.97	(44)	
(45)		RadStd	0.14	0.23	0.24	0.42	0.42	0.52	0.34	0.72	0.53	0.30	0.28	0.17	0.14	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.49	+0.63	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page